

Practice Test 1: The Special Senses

Multiple Choice

1. Which of the following is NOT a special sense?
 - a) Vision
 - b) Smell
 - c) Pain
 - d) Taste
 2. What type of receptor is involved in olfaction?
 - a) Mechanoreceptor
 - b) Chemoreceptor
 - c) Photoreceptor
 - d) Thermoreceptor
 3. The fovea centralis contains:
 - a) Only rods
 - b) Only cones
 - c) Both rods and cones
 - d) No photoreceptors
 4. Which cranial nerve is responsible for transmitting taste sensations from the anterior two-thirds of the tongue?
 - a) CN V (Trigeminal)
 - b) CN VII (Facial)
 - c) CN IX (Glossopharyngeal)
 - d) CN X (Vagus)
 5. What structure in the ear detects rotational movements?
 - a) Cochlea
 - b) Utricle
 - c) Semicircular canals
 - d) Sacculle
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True/False

6. The olfactory epithelium is located in the inferior part of the nasal cavity.
 7. Bitter tastes are detected by ion channels.
 8. Rods are more sensitive to light than cones.
 9. The auditory ossicles amplify sound vibrations in the middle ear.
 10. The macula lutea has the highest concentration of rods in the retina.
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Fill in the Blank

11. The _____ gland secretes mucus that dissolves odorant molecules in the nasal cavity.
 12. The _____ nerve transmits sensory information for hearing to the brain.
 13. The _____ is the transparent part of the fibrous tunic that refracts light entering the eye.
 14. _____ equilibrium detects linear movements, while _____ equilibrium detects rotational movements.
 15. The _____ is the structure in the cochlea where hair cells detect sound vibrations.
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Short Answer

16. Explain how cyclic AMP (cAMP) is involved in the process of olfaction.
 17. Describe the role of the Eustachian tube in the middle ear.
 18. What are the five primary tastes, and what types of molecules or ions activate each?
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Practice Test 2: Endocrine System

Multiple Choice

1. Which hormone is lipid-soluble?
 - a) Insulin
 - b) Epinephrine
 - c) Cortisol
 - d) Prolactin
2. The hypothalamus regulates the anterior pituitary through:
 - a) Neural impulses
 - b) Releasing and inhibiting hormones
 - c) Second messengers
 - d) Direct secretion of hormones into the bloodstream
3. Which hormone stimulates uterine contractions during childbirth?
 - a) Oxytocin
 - b) ADH
 - c) Prolactin
 - d) FSH
4. What is the primary function of calcitonin?
 - a) Increase blood calcium levels
 - b) Decrease blood calcium levels
 - c) Stimulate osteoclast activity
 - d) Promote calcium absorption in the intestines

5. The adrenal medulla secretes:
 - a) Aldosterone and cortisol
 - b) Epinephrine and norepinephrine
 - c) Androgens and glucocorticoids
 - d) T3 and T4
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True/False

6. Water-soluble hormones bind to intracellular receptors.
 7. The posterior pituitary produces oxytocin and ADH.
 8. The zona glomerulosa of the adrenal cortex secretes aldosterone.
 9. Insulin increases blood glucose levels.
 10. The parathyroid hormone (PTH) promotes calcium absorption in the intestines.
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Fill in the Blank

11. Lipid-soluble hormones require _____ proteins to travel in the bloodstream.
 12. The _____ portal system connects the hypothalamus to the anterior pituitary.
 13. _____ is the most abundant glucocorticoid secreted by the adrenal cortex.
 14. The _____ gland secretes melatonin, which regulates the body's biological clock.
 15. The _____ hormone decreases urine production and conserves water.
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Short Answer

16. Compare and contrast the mechanisms of action for lipid-soluble and water-soluble hormones.
 17. List the three zones of the adrenal cortex and the hormones they secrete.
 18. Explain the negative feedback mechanism regulating thyroid hormone secretion.
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Practice Test 3: Reproductive System

Multiple Choice

1. What is the function of Sertoli cells in the testes?
 - a) Produce testosterone
 - b) Form the blood-testis barrier
 - c) Store sperm
 - d) Secrete progesterone
2. During oogenesis, the secondary oocyte is arrested in which phase of meiosis until fertilization?

- a) Prophase I
 - b) Metaphase II
 - c) Anaphase I
 - d) Telophase II
3. Which hormone triggers ovulation?
- a) FSH
 - b) LH
 - c) Estrogen
 - d) Progesterone
4. The corpus luteum primarily secretes:
- a) Testosterone
 - b) Progesterone
 - c) FSH
 - d) Inhibin
5. What is the site of fertilization in the female reproductive system?
- a) Ovary
 - b) Uterus
 - c) Fallopian tube
 - d) Cervix
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True/False

- 6. Spermatogenesis occurs in the epididymis.
 - 7. The seminal vesicles produce an alkaline fluid that neutralizes vaginal acidity.
 - 8. The myometrium is the innermost layer of the uterus.
 - 9. The stratum basalis of the endometrium is shed during menstruation.
 - 10. Oogenesis begins before birth, but spermatogenesis begins at puberty.
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Fill in the Blank

- 11. The _____ hormone stimulates testosterone production in Leydig cells.
 - 12. The _____ is the funnel-shaped portion of the fallopian tube near the ovary.
 - 13. The _____ ligament anchors the ovary to the uterus.
 - 14. The _____ phase of the uterine cycle involves thickening of the endometrium in preparation for implantation.
 - 15. The _____ gland produces a milky, slightly acidic fluid that contributes to semen.
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Short Answer

16. Describe the pathway of sperm from production to ejaculation.
 17. What are the four phases of the female reproductive cycle, and what happens in each?
 18. Explain the hormonal changes that occur if fertilization does not take place.
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